

Statistical Modelling and Inference

Solutions

Practical 6: ANCOVA

Spring 2 2021

Contents

Load data	1
Task 1: Primary examination of the data	1
Task 2: ANOVA	4
Task 3: Identical regression lines	5
Task 4: Parallel regression lines	6
Task 5: Separate regression lines	8
Task 6: Model Selection	9
Task 7: Model contrasts (OPTIONAL)	10

In this practical, we are going to look at predicting the relationship between City Miles per Gallon and Engine Displacement, while taking into account the Drive (type).

Load data

The mpg data is included in the `tidyverse` package. To load the data:

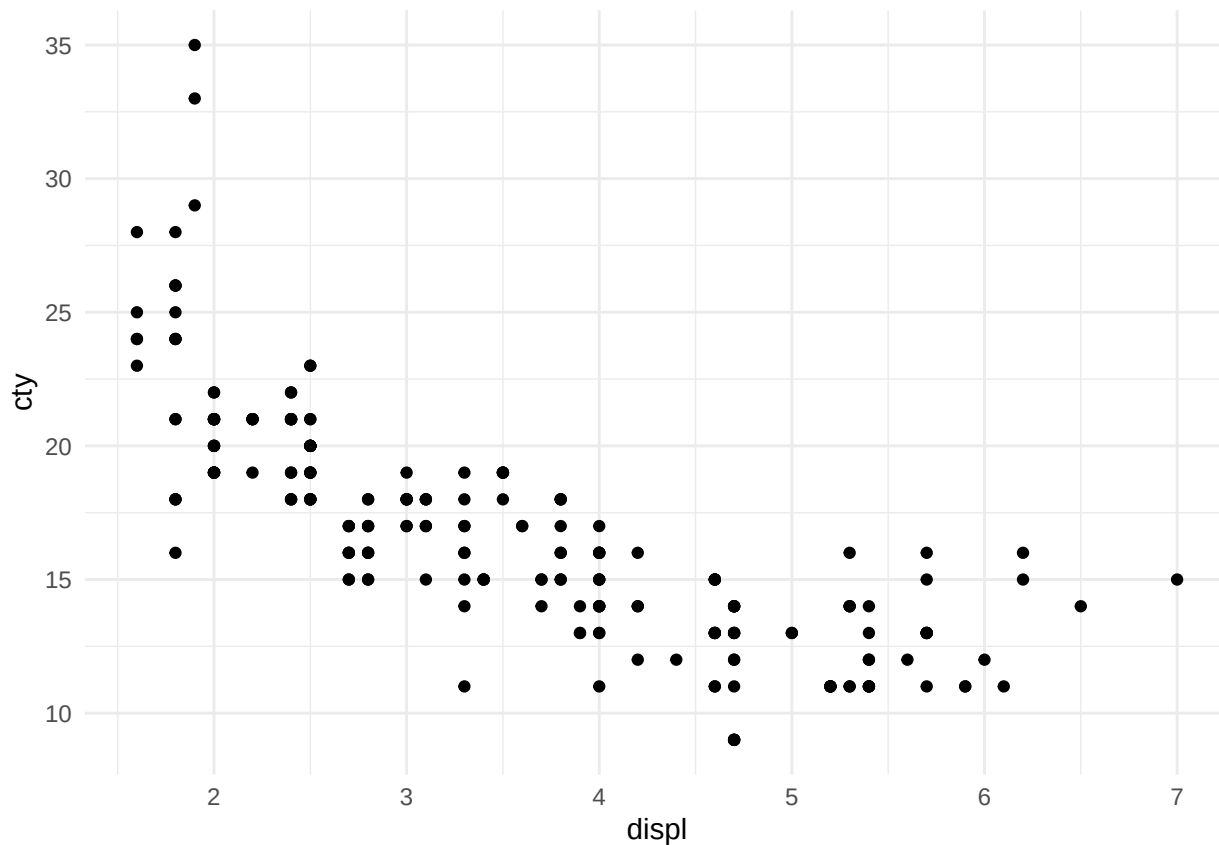
```
library(tidyverse)
data(mpg)
```

Task 1: Primary examination of the data

Quiz Questions

1. For the following scatter-plot, describe the relationship between Displacement and City Miles per Gallon.

```
mpg %>%
  ggplot(aes(displ, cty)) +
  geom_point()
```



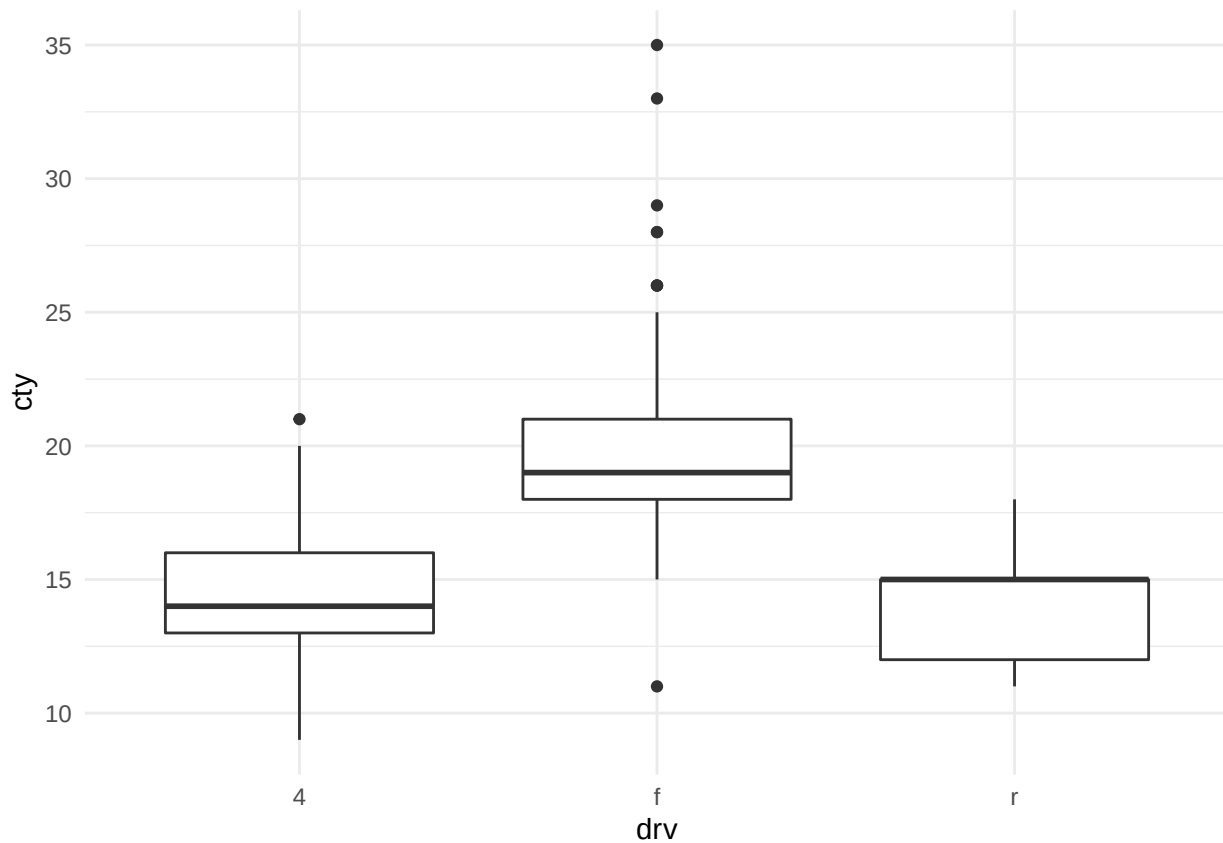
Solutions:

Overall, there is a negative strong non-linear relationship.

2. Produce a side-by-side box plot of `cty` for each level of `drv`. Does there appear to be a difference in City Miles per Gallon for each different Drive type?
-

Solutions:

```
mpg %>%  
  ggplot(aes(x = drv, y = cty)) +  
  geom_boxplot()
```

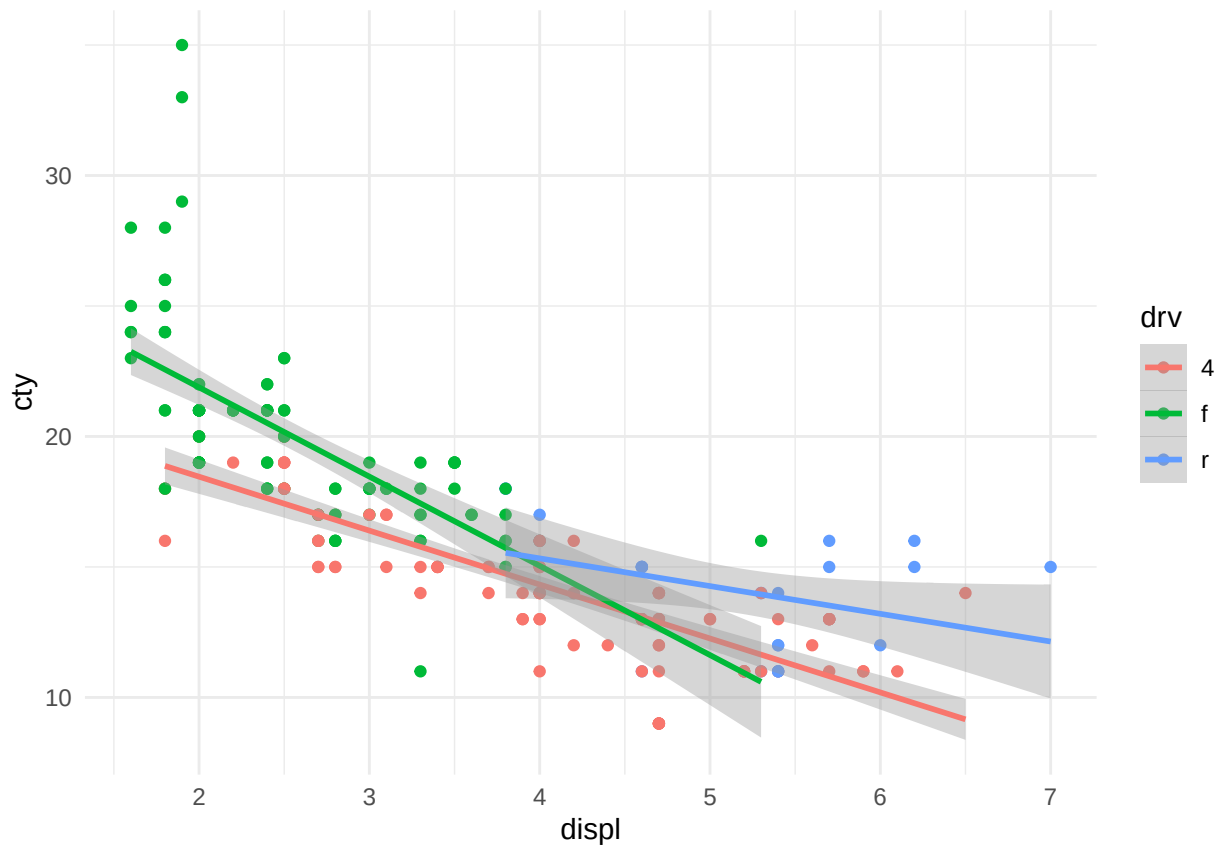


Solutions:

It seems that front wheel drives have a larger city miles per gallon usage on average.

3. Produce a scatter-plot of Displacement vs City miles per gallon with different colours and lines for each Drive (type). Describe the relationship between Displacement and City Miles per Gallon accounting for Drive (type).
4. Do you think that there is an interaction between Drive (type) and Displacement in predicting City Miles per Gallon?

```
mpg %>%
  ggplot(aes(displ, cty, col = drv)) +
  geom_point() +
  geom_smooth(method = "lm")
```



Solutions:

Once we account for drive we see that the 4-wheel drive and rear-wheel drive seems much more linear, while front-wheel drive still looks non-linear.

The lines for the three levels of drive appear to have different intercepts and slopes.

Task 2: ANOVA

Quiz Questions

5. By fitting the linear model $\text{cty} \sim \text{drv}$, perform an ANOVA to test the hypothesis

$$H_0 : \mu_f = \mu_r = \mu_4$$

where μ_f is the true mean cty for the front wheel drive cars, μ_r is the true mean cty for the rear wheel drive cars, and μ_4 is the true mean cty for the 4-wheel drive cars. You must report:

- The observed value of the test statistic
- The degrees of freedom for the reference F-distribution
- The associated p-value

Solutions:

```
mpg_anova_lm <- lm(cty ~ drv, data = mpg)
anova(mpg_anova_lm)

## Analysis of Variance Table
##
## Response: cty
##           Df Sum Sq Mean Sq F value    Pr(>F)
## drv         2 1878.8   939.41  92.676 < 2.2e-16 ***
## Residuals 231 2341.5    10.14
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Solutions:

- The observed value of the test statistic: 92.676
 - The degrees of freedom for the reference distribution: 2 and 231
 - The associated p-value: <0.001
-

Task 3: Identical regression lines

Quiz Questions

6. For the following observations, write down the design matrix for the identical regression of City Miles per Gallon on Displacement.

drv	displ	cty
4	1.8	18
f	1.8	18
r	5.3	14

Solutions:

```
mpg %>%
  select(drv, displ, cty) %>%
  group_by(drv) %>%
  slice(1) %>%
  model_matrix(cty ~ displ)
```

```
## # A tibble: 3 x 2
##   `(Intercept)` displ
##           <dbl> <dbl>
## 1             1   1.8
## 2             1   1.8
## 3             1   5.3
```

Quiz Questions

7. Fit a linear model for cty vs displ with identical regression lines for drv, that is, fit the model

$$cty_i = \beta_0 + \beta_1 displ_i + \varepsilon_i.$$

Using the model output, write down the line of best fit.

Solutions:

```
lm_identical <- lm(cty ~ displ, data = mpg)
summary(lm_identical)

##
## Call:
## lm(formula = cty ~ displ, data = mpg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.3109 -1.4695 -0.2566  1.1087 14.0064
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  25.9915     0.4821   53.91  <2e-16 ***
## displ       -2.6305     0.1302  -20.20  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.567 on 232 degrees of freedom
## Multiple R-squared:  0.6376, Adjusted R-squared:  0.6361
## F-statistic: 408.2 on 1 and 232 DF, p-value: < 2.2e-16
```

Solutions:

The line of best fit is:

$$cty = 25.99 - 2.63 \times displ.$$

Task 4: Parallel regression lines

Quiz Questions

8. For the following observations, write down the design matrix for the parallel regression of City Miles per Gallon on Displacement and Drive (type).

drv	displ	cty
4	1.8	18
f	1.8	18
r	5.3	14

Solutions:

```
mpg %>%
  select(drv, displ, cty) %>%
  group_by(drv) %>%
  slice(1) %>%
  model_matrix(cty ~ displ + drv)

## # A tibble: 3 x 4
##   `(Intercept)` displ  drv  drvr
##           <dbl> <dbl> <dbl> <dbl>
## 1             1   1.8     0     0
## 2             1   1.8     1     0
## 3             1   5.3     0     1
```

9. Fit a linear model for the parallel regression of City Miles per Gallon on Displacement and Drive (type), that is, fit the model

$$cty_i = \beta_0 + \beta_1 \text{displ}_i + \beta_2 \text{drv}(f)_i + \beta_3 \text{drv}(r)_i + \varepsilon_i,$$

where

$$\text{drv}(f)_i = \begin{cases} 1 & \text{if subject } i \text{ is a front-wheel drive,} \\ 0 & \text{otherwise,} \end{cases} \quad \text{drv}(r)_i = \begin{cases} 1 & \text{if subject } i \text{ is a rear-wheel drive,} \\ 0 & \text{otherwise.} \end{cases}$$

Using the model output, write down the line of best fit for each Drive (type).

Solutions:

```
lm_parallel <- lm(cty ~ displ + drv, data = mpg)
summary(lm_parallel)

##
## Call:
## lm(formula = cty ~ displ + drv, data = mpg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.2377 -1.3423 -0.1424  1.0129 13.4884
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  23.6796     0.6993   33.862 < 2e-16 ***
## displ       -2.3385     0.1652  -14.156 < 2e-16 ***
## drv         2.2752     0.4009   5.676 4.14e-08 ***
## drv         2.5045     0.5552   4.511 1.03e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.332 on 230 degrees of freedom
## Multiple R-squared:  0.7035, Adjusted R-squared:  0.6996
## F-statistic: 181.9 on 3 and 230 DF, p-value: < 2.2e-16
```

Solutions:
4-wheel drive

$$cty = 23.68 - 2.34 \times displ$$

front-wheel drive

$$cty = (23.68 + 2.28) - 2.34 \times displ = 25.95 - 2.34 \times displ.$$

rear-wheel drive

$$cty = (23.68 + 2.50) - 2.34 \times displ = 26.18 - 2.34 \times displ.$$

Task 5: Separate regression lines

Quiz Questions

10. For the following observations, write down the design matrix for the separate regression of City Miles per Gallon on Displacement and Drive.

drv	displ	cty
4	1.8	18
f	1.8	18
r	5.3	14

Solutions:

```
mpg %>%
  select(drv, displ, cty) %>%
  group_by(drv) %>%
  slice(1) %>%
  model_matrix(cty ~ displ * drv)
```

```
## # A tibble: 3 x 6
##   `(Intercept)` displ  drv  drv  `displ:drv` `displ:drv`
##         <dbl> <dbl> <dbl> <dbl>         <dbl>         <dbl>
## 1             1  1.8     0     0             0             0
## 2             1  1.8     1     0             1.8             0
## 3             1  5.3     0     1             0             5.3
```

11. Fit a linear model for the separate regression of City Miles per Gallon on Displacement and Drive, that is, fit the model

$$cty_i = \beta_0 + \beta_1 displ_i + \beta_2 \text{drv}(f)_i + \beta_3 \text{drv}(r)_i + \beta_4 displ_i \times \text{drv}(f)_i + \beta_5 displ_i \times \text{drv}(r)_i + \varepsilon_i,$$

where $\text{drv}(f)_i$ and $\text{drv}(r)_i$ are defined as above. Using the model output, write down the line of best fit for each Drive type.

Solutions:

```
lm_separate <- lm(cty ~ displ * drv, data = mpg)
summary(lm_separate)

##
## Call:
## lm(formula = cty ~ displ * drv, data = mpg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.4363 -1.2957 -0.0863  1.1203 12.7768
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  22.5914     0.8136   27.768 < 2e-16 ***
## displ       -2.0663     0.1958  -10.554 < 2e-16 ***
## drvf         6.1284     1.1632   5.269 3.18e-07 ***
## drvr        -3.0124     3.1043  -0.970 0.332872
## displ:drvf   -1.3529     0.3696  -3.661 0.000313 ***
## displ:drvr    1.0039     0.6048   1.660 0.098285 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.252 on 228 degrees of freedom
## Multiple R-squared:  0.7261, Adjusted R-squared:  0.7201
## F-statistic: 120.9 on 5 and 228 DF,  p-value: < 2.2e-16
```

Solutions:

4-wheel drive

$$cty = 22.59 - 2.07 \times displ$$

front-wheel drive

$$cty = (22.59 + 6.13) + (-2.07 - 1.35) \times displ = 28.72 - 3.42 \times displ.$$

rear-wheel drive

$$cty = (22.59 - 3.01) + (-2.07 + 1.00) \times displ = 19.58 - 1.06 \times displ.$$

Task 6: Model Selection

Quiz Questions

12. Use AIC or BIC to decide on the most appropriate model. Use your final model to predict the mean City Miles per Gallon for a four-wheel drive with a displacement of 4 litres.

Solutions:

```
AIC(lm_identical, lm_parallel, lm_separate)
```

```
##           df      AIC
## lm_identical  3 1109.336
## lm_parallel   5 1066.391
## lm_separate   7 1051.857
```

```
BIC(lm_identical, lm_parallel, lm_separate)
```

```
##           df      BIC
## lm_identical  3 1119.702
## lm_parallel   5 1083.667
## lm_separate   7 1076.045
```

Solutions:

From AIC (or BIC), the separate regression has the smallest AIC at 1051.857 (or BIC at 1076.045).

So the best model is the separate model.

```
predict(lm_separate, newdata=data.frame(drv="4", displ=4))
```

```
##           1
## 14.32608
```

Task 7: Model contrasts (OPTIONAL)

13. Calculate the 95% confidence interval for the **difference** in mean City Miles per Gallon for the rear-wheel and front-wheel drive (i.e. $\text{drv}(r) - \text{drv}(f)$) by:
- Loading the `emmeans` package.
 - Creating an `emmeans` object that looks at Drive type for the separate regressions model.
 - Creating a *contrast* to look at the hypothesis $\text{drv}(r) - \text{drv}(f)$ (Hint, what is your contrast vector going to be?)
 - Use the `confint` function to obtain your confidence intervals.
-

Solutions:

```
# using emmeans package
library(emmeans)
lm_emmeans <- emmeans(lm_separate, "drv")
lm_contrast <- contrast(lm_emmeans, method=list("drv.r-f"=c(0,-1,1)))
confint(lm_contrast)
```

```
## contrast estimate SE df lower.CL upper.CL
## drv.r-f      -0.958 1.13 228   -3.19    1.27
##
## Confidence level used: 0.95
```

```

## using formulae
X <- model.matrix(lm_separate)
displ_mean <- mean(mpg$displ)
lambda <- matrix(c(0, 0, -1, 1, -1 * displ_mean, displ_mean), ncol(X))
a <- t(lambda) %*% solve(t(X) %*% X) %*% lambda; a

##           [,1]
## [1,] 0.2531494

se <- summary(lm_separate)$sigma; se

## [1] 2.251718

lm_beta <- lm_separate$coefficients; lm_beta

## (Intercept)      displ      drv    drv    displ:drv    displ:drv
## 22.591424    -2.066335    6.128405   -3.012409   -1.352920    1.003929

cv <- qt(1-0.025, lm_separate$df.residual); cv

## [1] 1.970423

cv * se * sqrt(a)

##           [,1]
## [1,] 2.232348

se * sqrt(a)

##           [,1]
## [1,] 1.132928

sum(lambda * lm_beta)

## [1] -0.9583194

sum(lambda * lm_beta) + c(-1, 1) * as.numeric(cv * se * sqrt(a))

## [1] -3.190667  1.274029

```